1 **Design & Implementation of Human-Computer Interfaces**
NPTEL-MOOCS

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2 **L13 – Interface Design Guidelines II**

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4 **Recap**

- We are discussing interface design guidelines
  - Discussed one set of guidelines (Shneiderman's 8 golden rules)
- Today we shall introduce another guideline set (Norman's 7 principles)

5 **Background**

- The principles are based on a descriptive model of human-computer interaction
- Let's understand the model first

6 **Norman's Model of Interaction**

7 **Norman's Model of Interaction**

- Proposed by Norman [1988] - used extensively to represent user interaction with GUIs

8 **Norman's Model of Interaction**

- Represents behavior of a user of interactive systems in terms of a series of "actions"

9 **Norman's Model of Interaction**

- These actions represent the cognitive and sensory-motor activities of the user

10 **Norman's Model of Interaction**

- There are SEVEN such actions defined
- Consequently, the model is also known as the "seven stages of action"

11 **Two Stages**

- There are broadly two stages of interaction
  - "Execution" stage
  - "Evaluation" stage

12 **The Seven Actions**

13 **Execution Stage**
- In the execution stage, we carry out FOUR actions

14 **Execution Stage**
- The very first thing we do is to establish a goal
- For example, select a button on the screen

15 **Execution Stage**
- Once the goal is set, we formulate our intention to achieve the goal

16 **Execution Stage**
- Next, we translate our intention to sequence of real-world tasks
  - Ex - tasks required to select a button on the screen are: grab the mouse, drag the pointer to the button, and finally press the mouse button

17 **Execution Stage**
- Finally comes the actual execution of the task

18 **Execution Stage**
- The first three actions are purely mental
- The last action involves human motor action

19 **Evaluation Stage**
- In this stage, we first perceive the "system state" – i.e., the state of the interface
  - For example, after we perform the tasks to select a button, we look at the screen

20 **Evaluation Stage**
- Perception followed by interpretation; we try to make sense of the sensory inputs
  - E.g., if the color of the button changes after we select it, we might interpret this as "the button is selected"; otherwise, we might interpret as "button not selected"

21 **Evaluation Stage**
- Finally, we evaluate result of our actions by comparing with the goal state
  - For example, we may compare the interpretation that the "button is selected" to our goal state "select the button"

22 **Evaluation Stage**
- The first action in this stage is perception, related to our sensory organs (eye, ear etc.)
- The last two actions are purely mental (cognitive)

23 **The Execution-Evaluation Cycle**
- The model proposes to view any interaction as a cycle of these two stages (execution and evaluation)
  - If the evaluation matches with the goal, the cycle stops
  - Otherwise, it continues till the goal is achieved

24 **The Execution-Evaluation Cycle**

25 **Norman's Model - Use**

- Helped to develop two powerful concepts

26 **Norman's Model - Use**

- The gulf of execution - actions we specify to translate intentions may not be supported by the interface leading to gap (gulf) between the first three and the last actions in the execution stage

27 **Norman's Model - Use**

- The gulf of evaluation - Our interpretation of the interface state, based on our perception, may not match with the actual state of the interface (pointing to a gap or gulf between the first two and the last actions in the evaluation stage)

28 **Norman's Model - Use**

- These concepts actually were found to be quite useful in analyzing GUI and other interface designs and identifying design flaws
- Particularly in the context of errors during interaction (explained in terms of "slips and mistakes")

29 **Human Error: Slips and Mistakes**

- Slip
  - Understand system and goal
  - Correct formulation of action
  - Incorrect action

30 **Human Error: Slips and Mistakes**

- Mistake
  - May not even have right goal!

31 **Human Error: Slips and Mistakes**

- Fixing things?
  - Slip – better interface design
  - Mistake – better understanding of system

32 **Norman's Model - Use**

- Those were also used to recommend design guidelines

33 **Norman's 7 Principles**

34 **Norman's Seven Principles**

1. Use both knowledge in the world and knowledge in the head
2. Simplify the structure of tasks
3. Make things visible: bridge the gulfs of execution and evaluation
4. Get the mappings right
5. Exploit the power of constraints, both natural and artificial
6. Design for error
7. When all else fails, standardize

35 **Book**

- Bhattacharya, S. (2019). Human-Computer Interaction: User-Centric Computing for Design, McGraw-Hill India
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